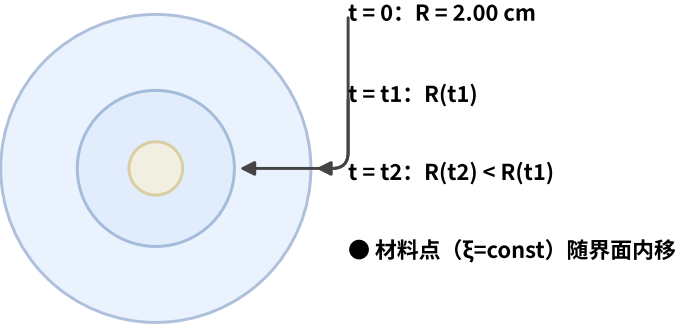


物理域（收缩）：药材截面 $r \in [0, R(t)]$



随体坐标变换
 $\xi = r / R(t) \in [0, 1]$
 $(1/r) \partial r (r \cdot a \cdot \partial r \phi) \rightarrow 1/(R^2 \xi) \cdot \partial \xi (\xi \cdot a \cdot \partial \xi \phi)$

固定计算域 $\xi \in [0, 1]$
节点 $\xi_j = j/N$ 恒定不动
物理位置 $r_j(t) = \xi_j \cdot R(t)$ 整体伸缩

边界条件不变： $\xi=0$ 对称 $\partial T / \partial \xi = \partial C / \partial \xi = 0$;
 $\xi=1$: $(k/R) \partial T / \partial \xi = h(T_a - T)$, $(D/R) \partial C / \partial \xi = -h_m(C - C_a)$